

Implementation of Trapezoidal Method in C++ Using $I = \int_0^2 e^{-x^2} dx$

Code:

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#include<bits/stdc++.h>
using namespace std;

double f(double x){
    return exp((-1)*x*x);
}

int main(){
    double a = 0, b = 2;
    double h, x, integral, sum;
    int n;
    cout << "Enter number of intervals: ";
    cin >> n;
    h = (b-a)/n;
    cout << endl;
    cout << " i      |      x      |      f(x)      |      I\n";
    cout << "-----\n";
    sum = 0;
    for(int i = 1; i < n; i++){
        x = a + ( i * h );
        sum = sum + 2 * f(x);
        integral = ( h/2 ) * (f(a) + sum + f(b));
        cout << fixed << setprecision(4) << " " << i << "      |      " << x
              << "      |      " << f(x) << "      |      " << integral << endl;
    }
    integral = ( h/2 ) * (f(a) + sum + f(b));
    cout << endl << "Final Integral = " << integral;
    return 0;
}
```

Output:

Enter number of intervals: 10

i	x	f(x)	I
1	0.2000	0.9608	0.2940
2	0.4000	0.8521	0.4644
3	0.6000	0.6977	0.6040
4	0.8000	0.5273	0.7094
5	1.0000	0.3679	0.7830
6	1.2000	0.2369	0.8304
7	1.4000	0.1409	0.8585
8	1.6000	0.0773	0.8740
9	1.8000	0.0392	0.8818

Final Integral = 0.8818